Hyperspectral Image Denoising via Double Subspace Deep Prior

Kexin Shi, Jiangjun Peng, *Member, IEEE*, Jing Gao, Yisi Luo, and Shuang Xu

***Abstract*— Hyperspectral image (HSI) denoising is an essential preprocessing step for downstream applications. Fully characterizing the spatial–spectral priors of HSI is crucial for HSI denoising tasks. In recent years, denoising methods based on low-rank subspaces have garnered attention. Within the low-rank decomposition framework (LRDF), the restoration of HSIs can be formulated as a problem of restoring two subspace factors. Since the rank of the HSI data has been predetermined by LRDF, subspace-based methods have already characterized the spectral low-rankness information. Next, subspace-based methods only need to encode spatial priors for HSIs. Existing subspace-based methods either rely on a manual-designed regularization or a pre-trained deep neural network. The former fails to fully capture the intrinsic priors of the HSI, while the latter may encounter generalization issues. Inspired by the unsupervised deep image prior (DIP) technique, this article proposes a double subspace deep prior (DSDP) model to track the mentioned issues. In this model, the two subspace factors are parallelly represented by two deep neural networks. By incorporating popular attention modules into classical convolutional neural networks, the well-designed subspace factor neural network can effectively capture the deep prior of the two subspace factors separately from each HSI in an unsupervised manner. Additionally, the total variation (TV) regularizer is added to constrain the generation of the subspace factor neural network, and further to ensure the effectiveness and robustness of the parameter learning process. Extensive experiments demonstrate that our method outperforms a series of competing methods.**

***Index Terms*— Attention module, convolutional neural network, subspace factor neural network, unsupervised deep image prior (DIP).**

Received 11 May 2024; revised 5 August 2024; accepted 2 September 2024. Date of publication 11 September 2024; date of current version 26 September 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFA1004100, in part by the National Natural Science Foundation of China under Grant 12201497 and Grant 12401674, in part by the Fundamental Research Funds for the Central Universities under Grant D5000240095, in part by Shaanxi Fundamental Science Research Project for Mathematics and Physics under Grant 22JSQ033, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515011358 and Grant 2024A1515011851, and in part by the Key Research and Development Plan of Shaanxi Province under Grant 2024GX-YBXM-145. *(Corresponding authors: Jiangjun Peng; Shuang Xu.)*

Kexin Shi, Jing Gao, and Yisi Luo are with the School of Mathematics and Statistics and Ministry of Education Key Laboratory of Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: kexinshi@stu.xjtu.edu.cn; jgao@mail.xjtu.edu.cn; yisiluo1221@foxmail.com).

Jiangjun Peng is with the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an 710021, China, and also with the Research and Development Institute, Northwestern Polytechnical University, Shenzhen 518063, China (e-mail: pengjj@nwpu.edu.cn).

Shuang Xu is with the Research and Development Institute, Northwestern Polytechnical University, Shenzhen 518063, China (e-mail: xs@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3457792

I. Introduction

HYPERSPECTRAL imaging (HSI) is an advanced imaging technique that captures images at hundreds of contiguous spectral bands across the electromagnetic spectrum. These images provide rich spectral–spatial information, making them valuable for various applications such as classification [1], [2], [3], [4], super-resolution [5], haze removal [6], compressive sensing [7], unmixing [8], and mineral exploration [9]. However, HSI data often suffers from noise due to various factors including sensor limitations, atmospheric conditions, and environmental interference [10]. Such noise can degrade the quality of spectral signatures and affect the effect of subsequent applications. Mining and utilizing prior information from HSI is crucial for HSI denoising. Based on the technique of mining priors, current HSI denoising methods can be categorized into two types: model-based and data-driven.

Model-based methods primarily construct regularization terms based on statistical approaches to encode prior information in HSI. The priors in HSI can be divided into two categories: spectral low-rankness [11], [12], [13], [14], [15], [16], [17] and spatial priors, where spatial priors mainly include local smoothness [18], [19], [20] and non-local-similarity [21], [22], [23]. The efficiency of model-based methods on raw data priors is generally low [24]. Hence, some fast model-based methods based on subspace prior mining have emerged [25], [26], [27]. The theoretical foundation of subspace-based fast algorithms originates from the LMM of HSI. Based on the concept of unmixing, the spectral curve of any point in space (representing a certain material) can be represented by a linear combination of the spectral curves for multiple materials (a set of bases). The spectral curves of multiple materials form an endmember matrix, while the combination coefficients of all points in space form an abundance map. If the number of bases in the endmember matrix is chosen to be the rank, the LMM corresponds precisely to the low-rank matrix/tensor decomposition (LRTD). The LMM framework is placed in Fig. 1. From Fig. 1(a), it can be seen that the mode-3 slice of the abundance map shares a similar texture structure as HSI, while the endmember vector (basis) in the endmember matrix exhibits continuity. The characteristics of subspace factors conveniently correspond to raw HSI data in the spatial image structure and spectral direction continuity; therefore, regularizer terms can be directly defined on smaller-sized subspace factors to encode the structural priors of HSI, thus achieving acceleration effects [27]. However, both raw space-based and subspace-based methods face the

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

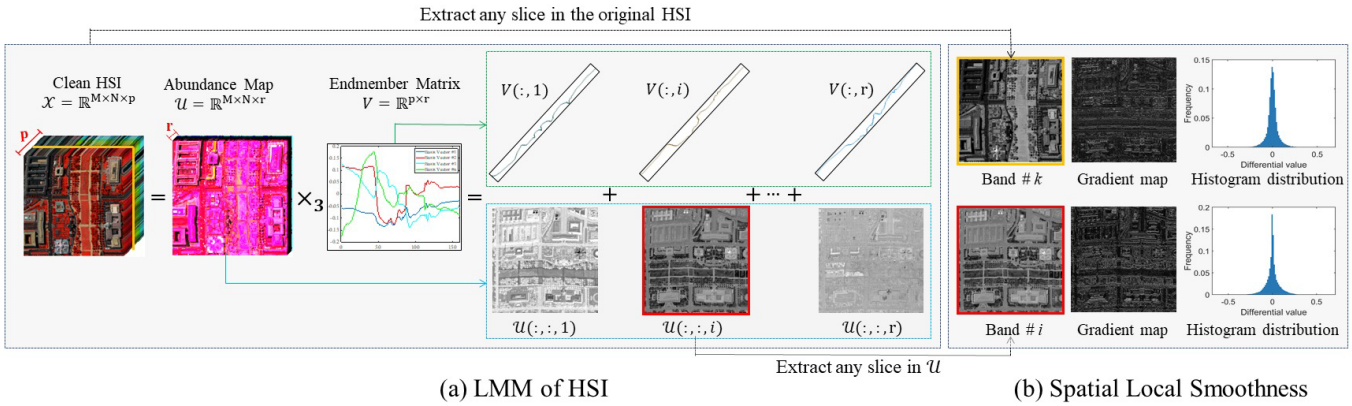


Fig. 1. LMM of HSI and its structure property analysis. (a) Linear mixing model (LMM) of HSI is represented as $\mathcal{X} = \mathcal{U} \times_3 \mathbf{V}$, where $\times_3$ is the mode-product between tensor and matrix along mode-3, $\mathcal{U}$ is an abundance map composed of a series of image slices $\mathcal{U}(:, :, i)$, and $\mathbf{V}$ is an endmember matrix composed of individual bases $\mathbf{V}(:, i)$, with the number of bases being $r(r \ll p)$. (b) Analyze the structural characteristics of a randomly selected image from the HSI and abundance map, and observe that both images exhibit strong sparsity in the gradient maps obtained after the differential operation (indicating local smoothness in the original images).

challenge of insufficient prior characterization, because manually defined regularizer terms may not adequately capture the unique prior information of the data [28].

Data-driven methods, on the other hand, directly leverage the powerful feature extraction capabilities of neural networks to learn the prior information from the HSI data. According to the training mode of the deep network, they can be broadly divided into three categories: supervised learning, plug-and-play, and unsupervised learning. The supervised learning mode is to directly learn a deep denoiser using paired clean-noisy data, which is intuitive and fast in inference [29], [30], [31], [32], [33], [34]. However, when encountering inconsistencies between testing and training data, the network's generalization performance sharply declines. The plug-and-play mode involves directly selecting a pre-trained network to act as a denoiser [35], [36]. However, since there may be structure mismatches and noise level discrepancies between the training domain selected by the pre-training model and the HSI image, there remains considerable room for improvement. The unsupervised learning mode directly uses a carefully designed network to denoise images from random input whiten-noise [37], [38], [39], that is, deep image prior (DIP) technique [40]. The basic principle behind this method is that convolutional neural networks have a good ability to encode image structure. However, due to the rich spectral and spatial information inherent in HSI, it is strenuous for a single network to effectively capture the spectral–spatial priors of HSIs in unsupervised mode. Besides, networks in unsupervised mode encounter the challenge of determining the optimal early stopping criteria to obtain the optimal solution.

To alleviate the shortcomings of the above model-based and data-driven methods and leverage the advantages of various methods, we turn our attention to mining subspace factor priors based on deep networks. For HSI, the most important priors are the *spectral correlation* and spatial structure property. From Fig. 1, we can see that the *spectral correlation* of the HSI can be encoded by LRTD, that is, $\mathcal{X} = \mathcal{U} \times_3 \mathbf{V}^T$, where $\mathcal{U}$ and $\mathbf{V}$ are called the abundance tensor map and base matrix, respectively. With LRTD, we transfer the problem of the raw HSI restoration into two subspace factors restoration. For subspace factors, continuity is a particularly important prior. Specifically, $\mathcal{U}$ has *spatial local smoothness*, and $\mathbf{V}$ has *spectral continuity* as shown in Fig. 1. Continuity is generally characterized by total variation (TV) regularizer. However, the TV regularizer fails to capture richer prior knowledge in the data, such as DIP, which has been shown to be very effective for HSI restoration in DHP [37] and S2DIP [41]. Besides, we further found that different slices of the abundance tensor $\mathcal{U}$ showed statistical correlation, that is, *factor correlation* as shown in Fig. 2. This inspires us to pay more attention to the information interaction between different slices of the abundance tensor $\mathcal{U}$ when modeling subspace factors, which is ignored by DS2DP [38]. Based on the above analysis, this article proposes an unsupervised model named double subspace deep prior (DSDP) based on the CNN and transformer modules and the idea of directly parameterizing the subspace factors as a whole. Compared with the previous DIP-based model, DSDP can better achieve the characterization of the HSI data prior. For easier comparison, we summarize the details of DSDP and other DIP methods in terms of prior encoding capabilities and modeling tools in Table I. From Table I, we can easily observe that only the DSDP model can encode all five properties. The whole process is placed in Fig. 3.

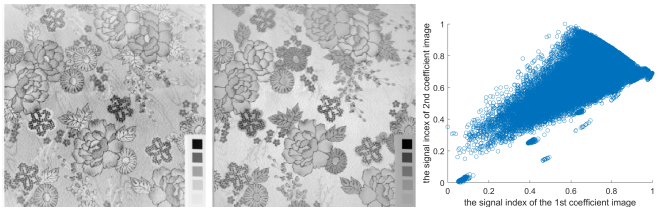


Fig. 2. Statistical correlations between Eigen images on *cloth* data in CAVE dataset. (a) First coefficient image. (b) Second coefficient image. (c) Scatter map of (a) and (b).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

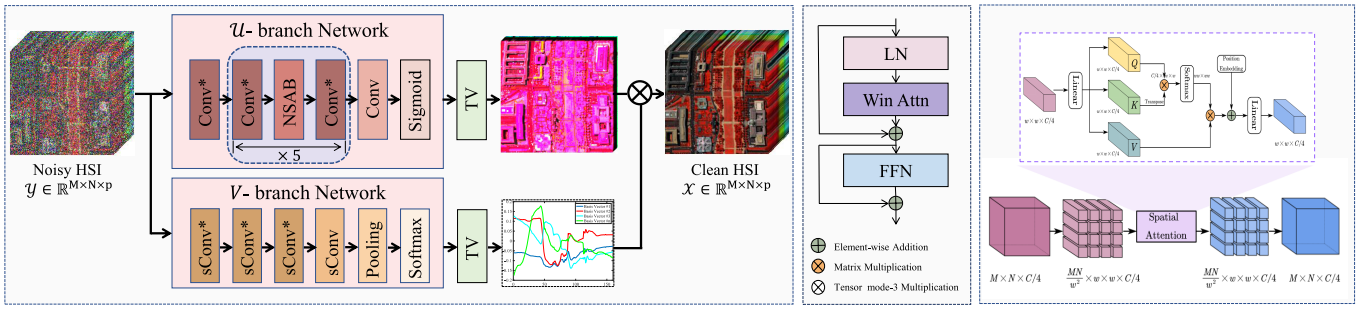


Fig. 3. Illustration of the proposed DSDP-TV model. (a) Overall architecture of DSDP-TV consists of a $\mathcal{U}$-branch network, **V**-branch networks, and the TV operator; Conv represents a convolutional layer, and SConv represents the convolutional layer with a stride of 2. Conv*/SConv* represents the convolutional layer with a batch-normalization layer and a leaky-ReLU layer. (b) NSAB module in the $\mathcal{U}$-branch network. (c) Window-based spatial attention operations in the NSAB module, and w represents the size of windows.

TABLE I
COMPARISON OF DIFFERENT DIP-BASED METHODS

Methods	Modeling Tools	Network Module	Deep image prior	Spectral correlation	Spatial local smoothness	Spectral continuity	Factor correlation
DHP	DIP	CNNs	✓	✗	✗	✗	✗
S2DIP	DIP + SSTV	CNNs	✓	✗	✓	✓	✗
DS2DP	DIP(S)[1]+ LRDF	CNNs	✓	✓	✗	✗	✗
DSDP	DIP(T)[2]+ LRDF + TV	CNNs +Transformer	✓	✓	✓	✓	✓

In summary, this work has the following contributions.

1) From a modeling perspective, the proposed DSDP transforms the network design from a single network to two networks under unsupervised mode. This allows DSDP to directly leverage the low-rankness of HSI spectral bands and promote the effective extraction of deep priors by the subspace networks. As a result, DSDP provides a more effective scheme for characterizing HSI priors.
2) From a network architecture design perspective, this article introduces subspace network $\mathcal{U}$-branch with joint multichannel convolution and spatial attention modules to enhance the characterization of spatial structure for coefficient maps, and subspace network **V**-branch with joint multichannel convolutional kernels and multibranch pooling to better represent the spectral characteristics for basis matrix.
3) From an optimization perspective, to improve the robustness of the unsupervised network-solving process, we integrate spatial local smoothness prior to coefficient representation and spectral local continuity prior to the basis matrix into the optimization problem using TV regularization, creating the DSDP-TV model. The local smoothness and continuity prior strengthen the robustness of the DSDP-TV model's optimization process in unsupervised mode.
4) From an experimental perspective, extensive experiments have demonstrated that the proposed DSDP-TV model outperforms state-of-the-art techniques.

[1]DIP(S) means separate DIP, which uses many parallel networks to represent each subspace factor.

[2]DIP(T) means together DIP, which uses an entire network to represent each subspace factor.

The remainder of this article is organized as follows. Section II reviews related works. Section III presents the DSDP-TV model and its optimization process. Section IV demonstrates the experimental results. Section V further represents some discussions about the experimental results. Conclusions are finally made in Section VI. Throughout the article, we denote scalars, vectors, matrices, and tensors in light, bold lower case letters, bold upper case letters, and upper cursive letters, respectively.

II. Related Work

In this section, we provide a brief overview of the current model-based approaches and several data-driven methods for HSI noise removal.

A. Model-Based Methods

Model-based methods focus on how to propose an appropriate regularizer to characterize the prior information of HSI data. Typical data priors include low-rank priors for spectral dimension as well as local smoothness and non-local self-similarity priors in the spatial domain. Low-rank prior refers to the existence of a low-dimensional subspace in HSI along the direction of the spectral band. Generally, it can be modeled by nuclear norm or LRMD/LRTD [18], [19], [23], [42], [43]. Typical methods based on nuclear norm include RPCA [11], WNNM [44], [45], TRPCA [13], [46]; and typical methods based on LRMD/LRTD that includes HyRes [47], LRMR [12], NMoG [48], and many others [43], [49]. Local smooth prior means that similar objects are usually distributed in local areas with high probability, so adjacent elements generally have the same or similar pixel values. Local smooth priors are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

generally characterized by TV regularizers [50], and are often embedded in low-rank decomposition frameworks (LRDFs) to obtain better repair performance. Typical methods include E-3DTV [15], LRTV [18], LRTDTV [19], and CTV [51]. Nonlocal spatial similarity means that there are always similar patterns in the spatial dimensions; thus this prior is often integrated into the LRMD/LRTD to eliminate Gaussian noise. Classic non-local-similarity-based denoising methods include KBR [21], BM4D [52], and LLRT [53]. Due to the huge size of HSIs, the regularizer term on the raw space is inefficient to solve. To alleviate this drawback, some subspace-based methods have been proposed. The low-rank nature of HSI has been encoded by a low-rank decomposition, while other characteristics of the HSI are inherited by the subspace factors, so the existing regular terms can be defined directly on the subspace factors. Since the size of the subspace factors is much smaller than raw HSI, the subspace-based methods have high solving efficiency. By mining the local smoothness and nonlocal self-similarity of subspace factors, a series of methods have been proposed, such as RCTV [27], NGmeet [26], FastHyDe [23], and LRTFDFR [54].

Model-based methods rely on the manual setting of regularizer terms with strong physical interpretability; thus, their inability to fully characterize the unique prior information of the data makes it still leave significant room for improvement in their denoising performance.

B. Data-Driven Methods

Unlike model-based methods, data-driven methods directly learn a denoising mapping from a large amount of data. These methods can be broadly divided into three categories: supervised learning, plug-and-play, and unsupervised learning. In supervised learning, deep denoisers are trained using paired clean-noisy data, such as in GRN [29], T3SC [34], QRNN3D [30], TRQ3DNet [55], HSID-CNN [31], HSI-DeNet [32], ADRN [56], 3-D-ADNet [57], MAC-Net [58], and SST [59]. While these methods achieve high fitting accuracy, they struggle with real-world test data due to limited generalization. The plug-and-play method directly applies a pre-trained network to act as a denoiser for a specific sub-problem in the denoising process. Due to the robustness demonstrated by the model toward the pre-trained network, the plug-and-play mode shows good denoising effects. Typical methods include FastHyMix [35] and AdHyDe [36]. However, since the selected network cannot well match data and different types of noise, there is still a lot of room for improvement.

The unsupervised learning directly uses a well-designed network to denoise from random white noise input. Examples include DHP [37], DS2DP [38], and S2DIP [41]. While these methods achieve denoising without labeled clean data, they often underperform compared to the other two types of data-driven methods. For instance, DHP [37] aims to capture spatial and spectral information of HSI, but it is challenging for the network to extract sufficient effective information in unsupervised mode. To enhance prior information extraction in unsupervised learning, a focus on both HSI structure modeling and suitable network module design is necessary.

III. Double Subspace Deep Prior

A. Problem Formulation

Let $\mathcal{X} \in \mathbb{R}^{M\times N\times p}$ denote a clean HSI, where $M \times N$ and p represent the spatial size and spectral bands, respectively. Under the assumption of universal additive noise, the observation model can be formalized as

$$\mathcal{Y} = \mathcal{X} + \mathcal{E} \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{M\times N\times p}$ indicates the observable degraded HSI and $\mathcal{E} \in \mathbb{R}^{M\times N\times p}$ indicates the mixture of various noises, including Gaussian noise, impulse noise, stripe noise, deadlines, and missing pixels [18], [19].

Recovering $\mathcal{X}$ from $\mathcal{Y}$ is a typical inverse problem, and solving inverse problems requires incorporating prior knowledge of the data. The typical HSI priors contain the low-rankness of spectra and spatial structural characteristics. As indicated by the LMM of HSI (see Fig. 1), the low-rankness of $\mathcal{X}$ can be characterized by low-rank tensor decomposition (i.e., $\mathcal{X} = \mathcal{U}\times_3\mathbf{V}$), and the prior information of $\mathcal{X}$ can be transferred to the prior information of the subspace factors. As is shown in Fig. 1, each slice in $\mathcal{U}$ possesses the similar spatial structure as $\mathcal{X}$, while each endmember vector in $\mathbf{V}$ exhibits local continuity in the spectral dimension of $\mathcal{X}$. Therefore, under the LRDF, based on subspace prior exploration, the denoising problem of HSI can be formulated as the optimization problem

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{U},\mathbf{V}} \text{loss}(\mathcal{X},\mathcal{Y}) + \alpha\mathcal{R}(\mathcal{U}) + \beta\mathcal{R}(\mathbf{V}) \\ &\text{s.t. } \mathcal{X} = \mathcal{U}\times_3\mathbf{V} \end{aligned} \tag{2}$$

where "loss" represents the loss term, $\mathcal{R}(\mathcal{U})$ and $\mathcal{R}(\mathbf{V})$ represent the regularizer terms to encode certain prior, and α and β are the trade-off parameters.

B. Parametric Representation

In model (2), how to mine the prior information of abundance maps and endmember matrices and encode them into $\mathcal{R}(\mathcal{U})$ and $\mathcal{R}(\mathbf{V})$ is the key to obtaining high-performance denoising performance. Existing methods usually focus on mining information from abundance maps. Common priors include local smoothness property [27] and non-local self-similarity [25]. In fact, the endmember matrix $\mathbf{V}$ also contains rich information. For example, NGmeet [26] and NS3R [60] simultaneously estimate the abundance map and the endmember matrix through refinement technology, which achieves very high-precision denoising effects.

In complex denoising scenarios, manually encoding data priors as regularizer terms cannot fully describe the unique prior contained in the data. Inspired by the idea of DIP [40], we turn to explore the deep prior behind the data, that is, utilizing a well-designed network to represent the prior of data. To reduce the burden on a single network in unsupervised mode as well as utilize decomposition factor correlations, we consider employing two pre-designed networks to parameterize the representation of the abundance map $\mathcal{U}$ and endmember matrix $\mathbf{V}$. In addition, we choose the noisy image $\mathcal{Y}$ as the input of the network because it contains more information about clean images than random

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

white noise. Meanwhile, considering the complexity of noise, we choose the more robust ℓ_1-norm [61] to measure the similarity between the clean image $\mathcal{X}$ to be restored and the noisy image $\mathcal{Y}$. Based on the above analysis, we obtain the following denoising model (3) as DSDP

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{U},\mathbf{V}} \|\mathcal{Y}-\mathcal{X}\|_1 \\ &\text{s.t. } \mathcal{X}=\mathcal{U}_{\times 3}\mathbf{V}^T \\ &\quad \mathcal{U}=g_u(\mathcal{Y};\theta_u) \\ &\quad \mathbf{V}=g_v(\mathcal{Y};\theta_v) \end{aligned} \tag{3}$$

where $g_u(\mathcal{Y};\theta_u)$ and $g_v(\mathcal{Y};\theta_v)$ are two deep networks, and θ_u and θ_v are the weight parameters of the networks correspondingly.

Remark 1: Compared to the basic subspace denoising model (2), although the DSDP model (3) does not explicitly specify the exact forms of $\mathcal{R}(\mathcal{U})$ and $\mathcal{R}(\mathbf{V})$ and encode them into the optimization objective, it still utilizes the prior information inherent in $\mathcal{U}$ and $\mathbf{V}$ on the process of network architecture design. In terms of purpose, setting $\mathcal{R}(\mathcal{U})$ and $\mathcal{R}(\mathbf{V})$ in model (2) aims to assist the model in obtaining better $\mathcal{U}$ and $\mathbf{V}$. In the optimization process of the DSDP model (3), abundance map $\mathcal{U}$ and endmember matrix $\mathbf{V}$ are parameterized as two neural networks. By setting appropriate network structures, the networks themselves can likewise utilize the prior information of the data to obtain better $\mathcal{U}$ and $\mathbf{V}$. Therefore, model (2) explicitly utilizes the prior information of the subspace, while DSDP model (3) implicitly utilizes the prior information of the subspace.

C. Factor Correlation and Network Architecture

To denoise using model (3), the key lies in the design of the network structure. From Fig. 2, the different feature maps exhibit apparent statistical correlation, while the basis vectors with no orthogonal constraints also exhibit correlation. The factors corrections inspire us to learn the abundance tensor and the base matrix cross channel, achieving potentially much better reconstruction. Therefore, we represent the entire $\mathcal{U}$ and $\mathbf{V}$ with a single sub-network separately, which is different from DS2DP [38]. Before analyzing the specific structure of the network, it is necessary to review the properties of the two subspace factors. As shown in Fig. 1, it is easily observed that the image slices in the abundance map $\mathcal{U}$ have obvious edge structures, and the structural texture looks alike across slices. Therefore, the spatial structure of the abundance map $\mathcal{U}$ can be effectively captured using traditional multichannel convolutional filters. Second, different regions of the images play different roles in the reconstruction of the image. Therefore, we consider using attention mechanisms to adaptively represent the importance of the different regions in the image. For these reasons, we ultimately constructed the $\mathcal{U}$-branch sub-network structure shown in Fig. 3. As seen in Fig. 1, it is easy to observe that a series of basis vectors in the endmember matrix $\mathbf{V}$ represent the spectral curves of multiple pure substances. To extract diverse spectral curves of multiple classes, it is better to consider the inner correlation of basis vectors and bundle all basis vectors together to learn as a whole. For this reason, we ultimately constructed the $\mathbf{V}$-branch sub-network structure shown in Fig. 3.

After some trials, the specific configuration of the two networks is set as follows.

1) *U-Branch:* A convolutional layer maps the channel of the input $\mathcal{Y}$ from p to C (C represents the number of convolutional filters, which is set as 256). Spatial feature extraction module repeatedly stacks five times to incorporate the complex information from local spatial regions, long-range context, and cross channels. Each spatial feature extraction module, respectively, contains a convolutional layer to reduce the channel into $C/4$, a non-local spatial attention block (NSAB) to obtain more similar or duplicate spatial information beyond neighbors, and another convolutional layer to return the channel into C. After that, a convolutional layer maps the channel of the feature map from C to r, and a sigmoid activate function is applied to guarantee the generated values ranging from 0 to 1. Finally, the output size of the $g_u(\mathcal{Y};\theta_u)$ is $M \times N \times r$.
2) *V-Branch:* Four layers of convolutional operators with a stride equal to 2 gradually down-sample the input with a size of $p \times M \times N$ to the latent variable with a size of $p \times M/16 \times N/16$. An adaptive average pooling layer reduces the spatial dimensions of the latent variable from $M/16 \times N/16$ into r, and a softmax function is inserted at the last layer to make sure the sum of each row is 1. Finally, the output size of the $g_v(\mathcal{Y};\theta_v)$ is $p \times r$.

Remark 2: The transformer modules play animportant role in DSDP by extracting spatial contextual information. Computing global spatial attention within pixels can extract more detailed information but cost more computational resources. As a substitute, window-based spatial attention balances computational complexity and representation performance under carefully chosen window size w. As shown in Fig. 3(b), we set the input feature map of NSAB as $\mathcal{F}_{\text{in}} \in \mathbb{R}^{M\times N\times C/4}$. Similar to the common transformer mode, the window-based spatial attention module with pre-LayerNorm is used to extract the spatial similarity. The feed forward networks (FFNs) combined with residual connections and LayerNorm layers are used to further enhance the representation ability. Then, we obtain the output of NSAB $\mathcal{F}_{\text{out}} \in \mathbb{R}^{M\times N\times C/4}$

$$\begin{aligned} \mathcal{O} &= \text{Win Attn}(\text{LN}(\mathcal{F}_{\text{in}})) \\ \mathcal{F}_{\text{out}} &= (\mathcal{F}_{\text{in}}+\mathcal{O})+\text{FFN}((\mathcal{F}_{\text{in}}+\mathcal{O})). \end{aligned} \tag{4}$$

About the window-based spatial attention module, that is, Fig. 3(c), the input feature maps are divided into non-overlapping patches. Integrated and reshaped, these patches utilize multihead self-attention to extract the spatial similarity beyond the neighbor's region. After linear weighting, the split patches are placed back to their original positions to get $\mathcal{O}$.

Remark 3: Although DSDP and DS2DP both use dual-branch networks to encode the deep prior of subspace factors under LRDF, there is a fundamental difference in their implementation. DS2DP models each slice of the abundance tensor and each basis vector of the base matrix with an independent network, which prevents fully leveraging the factor correlation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

In contrast, DSDP models the entire abundance tensor $\mathcal{U}$ with a single sub-network and the entire base matrix $\mathbf{V}$ with another single sub-network. This allows for deep feature interaction between different slices or basis vector, leading to a better parameterization of the abundance tensor and the base matrix. Additionally, we introduced a transformer architecture in our network design, which enhances the network's ability to represent DIPs.

D. Proposed Model

Given that solving $\mathcal{U}$ and $\mathbf{V}$ in the model (3) through the unsupervised manner may face a lack of robustness as other unsupervised networks [38], [41], here, we further consider adding some additional prior constraints to $\mathcal{U}$ and $\mathbf{V}$ to enhance the robustness of network.

Among all priors, local smoothness and local continuity are the most common types, which describes the statistical regularity that neighboring pixels tend to have similar grayscale values due to the high probability of being composed of the same class of substance. As seen in Fig. 1, we can easily find that each slice of the abundance map shows local smoothness, and each endmember vector of the endmember matrix shows local continuity in the direction of the spectral band. The local smoothness/continuity of the data can be converted into the sparsity of the gradient domain after the differential operation. For example, in Fig. 1(b), the local smoothness of the abundance map slice is converted into the sparsity of its gradient map. Therefore, the local smoothness/continuity of data $\mathbf{X}$ is generally encoded by the following TV regularizer:

$$\|\mathbf{X}\|_{\mathrm{TV}} = \|\nabla(\mathbf{X})\|_1 \tag{5}$$

where $\nabla(\cdot)$ and $\|\cdot\|_1$ represent the difference operator and ℓ_1-norm, respectively. For data with local smoothness/continuity in multiple directions, the difference operator of the above formula should be extended to the difference operator in the corresponding direction. For example, the abundance map $\mathcal{U}$ has local smoothness among the horizontal and vertical direction; then TV regularity of the abundance map $\mathcal{U}$ can be expressed as

$$\|\mathcal{U}\|_{\mathrm{TV}} = \|\nabla_h \mathcal{U}\|_1 + \|\nabla_v \mathcal{U}\|_1 \tag{6}$$

where $\nabla_h(\cdot)$ and $\nabla_v(\cdot)$ denote the first-order forward finite-difference operator among the horizontal and vertical direction, respectively. Similarly, $\mathbf{V}$ has only one dimension of local continuity along the spectral direction; then TV regularity of the abundance map $\mathbf{V}$ can be expressed as

$$\|\mathbf{V}\|_{\mathrm{TV}} = \|\nabla(\mathbf{V})\|_1. \tag{7}$$

Next, we add TV regularity of $\mathcal{U}$ and $\mathbf{V}$ to the DSDP model (3) to increase the robustness of the model, and then we can obtain the following DSDP-TV model (8):

$$\begin{aligned} \min_{\mathcal{X},\mathcal{U},\mathbf{V}} \ & \|\mathcal{Y} - \mathcal{X}\|_1 + \alpha\|\mathcal{U}\|_{\mathrm{TV}} + \beta\|\mathbf{V}\|_{\mathrm{TV}} \\ \text{s.t. } & \mathcal{X} = \mathcal{U} \times_3 \mathbf{V}^T \\ & \mathcal{U} = g_u(\mathcal{Y}; \theta_u) \\ & \mathbf{V} = g_v(\mathcal{Y}; \theta_v) \end{aligned} \tag{8}$$

Algorithm 1 DSDP-TV for HSI Denoising

Input: Observed image $\mathcal{Y}$; smoothness degree $\gamma = 0.8$; maximum iteration number T_max $= 2000$; relative error tolerance $\epsilon = 0.01$.

Initialization: Set iteration number $t = 1$; randomly initialize network parameters $g_u(\cdot; \theta_u^{(0)})$, $g_v(\cdot; \theta_v^{(0)})$; select suitable hyper parameters α, β.

1: **while** $t \le$ T_max**and**RelErr$^{(t)} < \epsilon$ **do**
2: Calculate the generation of subspace factor neural networks $\mathcal{U}^{(t-1)} = g_u(\mathcal{Y}; \theta_u^{(t-1)})$, $\mathbf{V}^{(t-1)} = g_v(\mathcal{Y}; \theta_v^{(t-1)})$;
3: Calculate the TV regularization terms $\alpha\|\mathcal{U}^{(t-1)}\|_{\mathrm{TV}} + \beta\|\mathbf{V}^{(t-1)}\|_{\mathrm{TV}}$;
4: Update the parameters $\theta_u^{(t-1)} \to \theta_u^{(t)}, \theta_v^{(t-1)} \to \theta_v^{(t)}$ by optimizing (9);
5: Check the relative error RelErr$^{(t)} := \frac{\|\mathrm{out}^{(t)} - \mathrm{out}^{(t-1)}\|_F^2}{\|\mathrm{out}^{(t-1)}\|_F^2}$;
6: Update t $t \to t+1$;
7: **end while**

Output: The final smoothed restored image out*.

where α and β are two trade-off parameters to control the influence of TV regularity. If α and β are set as zero, then DSDP-TV model (8) can be degenerated to DSDP model (3).

E. Optimization Process

The DSDP-TV model (8) is an unsupervised model with unknown network parameters. To facilitate the solution, we first bring the specific parametric expressions of $\mathcal{U}$ and $\mathbf{V}$ into the optimization objective terms of the model. Then, the obtained optimization objective of DSDP-TV model is as follows:

$$\begin{aligned} \mathcal{L}(\theta_u, \theta_v) := & \|\mathcal{Y} - g_u(\mathcal{Y}; \theta_u) \times_3 g_v(\mathcal{Y}; \theta_v)^T\|_1 \\ & + \alpha\|g_u(\mathcal{Y}; \theta_u)\|_{\mathrm{TV}} + \beta\|g_v(\mathcal{Y}; \theta_v)\|_{\mathrm{TV}}. \end{aligned} \tag{9}$$

For (9), the objective function only contains learnable network parameters θ_u and θ_v. This is a typical unconstrained minimization problem, so the generic gradient descent method can be used to update network parameters. In the process of training, the generation of the subspace factor networks in iteration t can be expressed as

$$\mathcal{X}^{(t)} = g_u\left(\mathcal{Y}; \theta_u^{(t)}\right) \times_3 g_v\left(\mathcal{Y}; \theta_v^{(t)}\right)^T. \tag{10}$$

We use the moving average to provide a smoother restored HSI by setting the ultimate estimation in iteration t as

$$\mathrm{out}^{(t)} = \gamma\mathrm{out}^{(t-1)} + (1-\gamma)\mathcal{X}^{(t)} \tag{11}$$

where γ indicates the degree of smoothness for network estimate, which is simply set as 0.8, and out$^{(t)}$ is the tth step estimation.

The workflow of the algorithm is summarized in Algorithm 1. After reaching the convergence condition, the network achieves optimal parameters $\{\theta_u^*, \theta_v^*\}$, and the final restored HSI is noted as out*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

IV. Experiments

In this section, we conduct sufficient experiments on both semi-real noise data and real noise data to more intuitively demonstrate the effectiveness of the proposed model.

Fourteen SOTA competing methods on both synthetic and real noise HSI data are implemented to evaluate the performance of the proposed model. They can be divided into two categories: eight model-based methods and six data-driven methods, that is, low-rank regularity methods (LRMR [12] and NMoG [48]), low-rank regularity combined with TV regularity methods (LRTV [18], LRTDTV [19], E-3DTV [15], and RCTV [27]), low-rank regularity combined with non-local self-similarity regularity methods (NGmeet [26] and GLF [23]), plug-and-play mode method (FastHyMix [35]), unsupervised mode methods (DHP [37], S2DIP [41], and DS2DP [38]), and supervised mode methods (HSID-CNN [31] and T3SC [34]). All the relevant parameters of the competing methods are initially configured based on their original settings or fine-tuned by the best MPSNR to guarantee their deserved performance. The results of the supervised learning methods HSID-CNN and T3SC are obtained by inferring in the pre-trained model on ICVL[3] dataset.

We adopt sub-images of HSI/MSI due to limited computational resources. For simulation experiments, *PaviaU*[4] dataset clipped into sizes of $200 \times 200 \times 93$ and *balloons*[5] dataset clipped into sizes of $256 \times 256 \times 31$, are used to generate semi-real noisy images. For real experiments, *GF-Shanghai*[6] dataset with size of $300 \times 300 \times 155$ and *Indian pines*[7] dataset with sizes of $145 \times 145 \times 210$ serve as real noise images. Besides, the values of the single image for each band are re-scaled within [0, 1] before conducting experiments.

All the experiments are conducted on a PC with Intel Core i5-8500 CPU @ 3.00 GHz, NVIDIA RTX 2080 Ti GPU, and 32 GB RAM.

A. Synthetic Data Experiments

The simulation experiments assess the performance of the proposed methods against competing methods in terms of quantitative quality and visual perception. The clean HSI dataset *PaviaU* and MSI dataset *balloons* serve as ground truth to generate noisy scenarios and evaluate denoising performance. We utilize three common quantitative quality metrics: MPSNR, MSSIM, and ERGAS. MPSNR and MSSIM represent the mean values of PSNR and SSIM across each band, respectively. Higher MPSNR and MSSIM values, along with lower ERGAS values, indicate greater improved image restoration. Additionally, we visualize spatial contextual preservation and spectral curve recovery for a more comprehensive evaluation.

[3]ICVL dataset available at http://icvl.cs.bgu.ac.il

[4]*PaviaU* dataset available at https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[5]*balloons* dataset is one of the MSI image in CAVE dataset, and CAVE dataset is available at https://www.cs.columbia.edu/CAVE/databases/multispectral/

[6]*GF-Shanghai* dataset is captured by the AHSI imager onboard Gaofen-5 satellites, which covers the suburbs of Shanghai City, China.

[7]*Indian pines* dataset available at https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

It is widely recognized that real HSI is often polluted by complex noise. In addition to the trivial iid Gaussian noise, non-iid Gaussian noise, impulse noise, stripes, and deadlines are also presented in real noise scenarios frequently. Thus, we consider six various noise cases by adding them to clean images to simulate real noise cases, shown as follows.

1) Each band of data is corrupted by zero-mean Gaussian noise with the same standard variance, which is 0.2.
2) Each band of data is corrupted by zero-mean Gaussian noise with different standard variances, which are randomly chosen from 0 to 0.4.
3) On the basis of case (a), each band of data is corrupted by impulse noise with a ratio of 0.2.
4) On the basis of case (b), each band of data is corrupted by impulse noise with the ratio randomly chosen from 0 to 0.4.
5) On the basis of case (d), 0%–45% of the bands are randomly selected to add stripe noise, where the number of strips is generated from 20 to 40 under the fixed width of 1.
6) On the basis of case (e), 30%–60% of the bands are randomly selected to add deadlines, where the number of deadlines is generated from 20 to 50 under the fixed width of 1.

1) Quantitative Comparison: Tables II and III list the quantitative results of competing methods on HSI/MSI in cases (a)–(f). To start, we acknowledge that our method does not perform as well as advanced non-local self-similarity-based methods (such as NGmeet, GLF, and FastHyMix) when handling Gaussian noise [cases (a) and (b)]. Although we cannot outperform NGmeet in removing Gaussian noise due to its exceptional ability to identify consistent patterns within grouping patches, our approach still achieves higher MPSNR values compared to most methods based on low-rank prior, local smoothness prior, and supervised mode. However, when it comes to complex noise or mixed noise, such as impulse, stripes, and deadlines [cases (c)–(f)], which are more common and realistic scenarios in HSI analysis, non-local self-similarity based methods cannot perform consistently superior for they fail to balance between removing the complex noises and preserving the structure and details. While some TV-based methods begin to stand out, that is, LRTV, LRTDTV, E-3DTV, and RCTV. In these scenarios, our method demonstrates state-of-the-art performance among various comparative methods, exhibiting consistent MPSNR gains, especially compared to similar unsupervised learning-based approaches such as DHP, S2DIP, and DS2DP. Moreover, we find that the supervised learning-based methods, HSID-CNN and T3SC, do not achieve top performance in any noise setting due to limited generation. Nonetheless, their denoising results are acceptable due to their remarkably fast inference times, showcasing the great potential of supervised methods. Finally, it should be noted that, despite belonging to the data-driven denoising methods, DSDP-TV cannot show comparable denoising time to FastHyMix, HSID-CNN, and T3SC. DSDP-TV method is committed to learning a proprietary denoiser for each noisy image from a randomly initialized network (self-supervised learning mode), while HSID-CNN and T3SC methods learn a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON OF ALL COMPETING METHODS ON *PaviaU* DATASET. THE BEST AND THE SECOND BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND *Italic Underline*, RESPECTIVELY

Noise Types	Metrics	Noisy	Model-based Methods								Data-driven Methods						
			NMoG	LRMR	LRTV	LRTDTV	E-3DTV	RCTV	NGmeet	GLF	FastHyMix	DHP	S2DIP	DS2DP	HSID-CNN	T3SC	DSDP-TV
Case (a)	MPSNR	13.97	29.34	29.06	31.57	31.59	30.38	32.13	**34.52**	*34.00*	33.63	31.02	33.05	31.38	25.30	31.36	33.33
	MSSIM	0.156	0.781	0.790	0.868	0.865	0.851	0.885	**0.938**	*0.931*	0.924	0.879	0.913	0.872	0.644	0.883	0.921
	ERGAS	729.5	125.4	127.4	118.3	95.14	114.0	89.10	**67.27**	*71.50*	74.83	100.7	79.62	97.19	194.1	118.3	77.25
Case (b)	MPSNR	16.65	38.33	30.34	33.96	32.99	32.38	30.65	30.71	**40.65**	*40.27*	30.12	34.44	33.29	25.68	30.43	36.08
	MSSIM	0.263	0.958	0.828	0.914	0.894	0.902	0.841	0.885	**0.982**	*0.979*	0.850	0.936	0.920	0.660	0.857	0.957
	ERGAS	864.9	53.90	113.8	180.4	93.69	97.83	132.1	107.1	**37.50**	*39.59*	113.4	69.13	79.43	196.5	125.8	57.07
Case (c)	MPSNR	10.15	27.04	27.45	30.41	30.35	29.30	30.40	22.90	23.69	23.76	23.21	25.54	*30.51*	24.63	23.00	**32.11**
	MSSIM	0.075	0.724	0.724	0.844	0.835	0.825	0.845	0.702	0.776	0.786	0.723	0.701	*0.851*	0.616	0.707	**0.898**
	ERGAS	1144.	161.8	152.6	156.2	110.5	128.3	109.0	265.7	247.4	246.7	260.0	189.3	*107.3*	210.0	277.1	**88.88**
Case (d)	MPSNR	10.53	32.22	28.84	31.45	31.75	31.34	31.89	23.61	25.31	26.69	24.15	27.05	*32.45*	24.50	23.80	**34.49**
	MSSIM	0.096	0.885	0.780	0.869	0.872	0.880	0.871	0.750	0.813	0.851	0.734	0.778	*0.901*	0.607	0.723	**0.938**
	ERGAS	1172.	221.5	132.3	216.0	106.3	103.8	114.2	272.5	251.4	243.9	271.7	177.7	*86.23*	220.2	283.5	**67.87**
Case (e)	MPSNR	9.753	*32.10*	28.01	31.38	31.77	31.51	31.30	21.43	22.29	23.74	21.13	26.31	31.67	24.05	21.68	**34.17**
	MSSIM	0.094	0.899	0.775	0.878	0.877	0.887	0.876	0.663	0.696	0.742	0.604	0.773	*0.903*	0.595	0.661	**0.940**
	ERGAS	1274.	249.7	155.2	236.2	116.8	103.0	117.6	362.3	332.4	331.8	365.7	202.2	*99.28*	233.6	349.2	**72.53**
Case (f)	MPSNR	10.47	15.82	25.95	27.71	28.43	28.09	28.38	22.83	24.11	25.02	22.84	25.06	*28.78*	23.99	23.23	**30.65**
	MSSIM	0.088	0.259	0.745	0.806	0.829	0.837	0.805	0.708	0.771	0.792	0.668	0.697	*0.839*	0.583	0.692	**0.886**
	ERGAS	1170.	585.9	215.1	296.0	169.6	**154.6**	218.5	281.1	256.1	256.8	287.4	219.5	195.3	234.7	283.7	*161.5*
Avergae Times(s)		-	50.79	137.8	66.00	78.66	17.93	3.260	50.63	308.7	**0.316**	2240.	21857.	4750.	*2.025*	9.824	690.3

TABLE III

QUANTITATIVE COMPARISON OF ALL COMPETING METHODS ON BALLOONS DATASET. THE BEST AND THE SECOND BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND *Italic Underline*, RESPECTIVELY

Noise Types	Metrics	Noisy	Model-based Methods								Data-driven Methods						
			NMoG	LRMR	LRTV	LRTDTV	E-3DTV	RCTV	NGmeet	GLF	FastHyMix	DHP	S2DIP	DS2DP	HSID-CNN	T3SC	DSDP-TV
Case(a)	MPSNR	13.98	25.46	27.09	32.66	34.21	31.54	37.63	*40.40*	35.43	39.11	38.62	**40.58**	35.96	34.08	32.46	36.33
	MSSIM	0.036	0.357	0.435	0.935	0.934	0.945	0.961	**0.982**	0.881	0.964	0.951	*0.971*	0.945	0.919	0.918	0.967
	ERGAS	701.3	188.3	155.2	84.86	71.22	111.0	46.50	*34.64*	137.3	40.24	41.08	**33.06**	58.44	69.24	115.5	58.01
Case(b)	MPSNR	16.72	29.40	27.20	37.45	36.70	34.25	35.17	32.20	37.93	39.11	38.10	**41.95**	37.77	34.98	32.29	*39.93*
	MSSIM	0.132	0.702	0.474	0.960	0.956	0.964	0.952	0.947	0.940	0.964	0.943	*0.977*	0.969	0.923	0.912	**0.983**
	ERGAS	758.4	397.4	251.8	51.14	60.37	95.22	64.49	89.81	112.2	*40.24*	45.21	**29.61**	51.85	69.72	115.8	41.38
Case(c)	MPSNR	10.05	22.64	25.08	31.18	33.11	30.85	*35.41*	23.75	23.16	23.95	24.01	30.28	34.03	33.16	22.66	**37.37**
	MSSIM	0.016	0.263	0.349	0.895	0.912	0.928	*0.937*	0.823	0.681	0.815	0.712	0.911	0.929	0.909	0.815	**0.963**
	ERGAS	1104.	264.5	195.2	102.7	80.82	117.9	*59.90*	229.6	304.1	224.5	222.1	108.1	71.15	77.10	274.3	**49.04**
Case(d)	MPSNR	10.44	25.90	24.73	34.25	34.54	32.45	35.53	23.90	24.97	26.98	26.23	32.25	*35.78*	32.51	23.59	**36.23**
	MSSIM	0.023	0.503	0.367	0.924	0.939	0.949	*0.945*	0.830	0.722	0.847	0.805	0.921	0.941	0.882	0.831	**0.969**
	ERGAS	1135	252.9	269.6	73.57	73.60	106.8	66.47	236.6	320.4	223.6	231.6	122.7	*60.28*	88.52	270.7	**58.10**
Case(e)	MPSNR	10.53	24.26	24.57	*34.73*	34.24	31.99	34.22	22.91	24.54	24.91	25.08	31.30	34.41	32.13	22.91	**35.88**
	MSSIM	0.031	0.421	0.380	*0.933*	0.921	0.939	0.932	0.781	0.720	0.633	0.785	0.909	0.917	0.875	0.807	**0.966**
	ERGAS	1174.	535.2	358.9	*70.39*	75.46	113.0	81.46	300.3	394.8	307.8	281.9	161.6	72.95	96.53	307.2	**62.12**
Case(f)	MPSNR	9.090	21.37	21.98	29.62	30.23	29.67	29.27	22.22	22.57	23.60	23.30	27.51	30.00	*30.74*	21.96	**31.53**
	MSSIM	0.015	0.334	0.299	0.879	*0.883*	**0.930**	0.879	0.786	0.671	0.734	0.785	0.873	0.877	0.854	0.797	0.864
	ERGAS	1288.	512.0	346.5	167.8	*138.4*	159.4	174.0	288.8	377.6	273.9	272.5	206.1	158.4	**110.1**	304.5	153.1
Average Times(s)		-	60.69	106.7	37.27	33.09	10.40	2.502	78.85	803.7	**0.596**	1108.	7106.	691.2	*1.638*	7.671	747.4

general denoiser from a large amount of training data and then directly generalize to test data (supervised learning mode), while FastHyMix adopts a pre-trained denoiser to solve a denoise sub-problem (plug-and-play mode). Despite achieving fast denoising, plug-and-play and supervised methods always demonstrate poor generalization performance on inconsistent testing data, which can be avoided for self-supervised methods. Although not achieving optimal denoising time, our denoising time is still competitive with similar self-supervised denoising methods DHP, S2DIP, and DS2DP due to our subspace modeling method and lightweight network structure design.

2) Visual Comparison: We display pseudo-RGB images of the recovered HSI/MSI of all the competing methods in case (f) as Figs. 4 and 6 to demonstrate the superiority of our method on spatially single-image denoising. The restored image is claimed better if it is more consistent with ground truth in spatial structure, texture information, and color fidelity detail. In each subfigure, the same subregion is marked and then enlarged three times for better visual perception. From these restored images, we have some findings: first, all of the methods are capable of removing complex mixed noise. Second, in addition to the performance metrics, our method also achieves impressive visual observation. It not only removes the impulse noise, stripe noise, and deadline contained in the degraded HSI but also preserves the spatial texture structure of the original HSI, with fewer artifacts or color deviations.

In addition, we also present the spectral curve recovery at a chosen location for all competing methods in case (f) in Figs. 5 and 7. This highlights our method's superiority in restoring spectral band signals. The methods that closely match the variation trend of the ground truth are deemed to better preserve the spectral structure. Our approach effectively maintains the redundant nonlinear spectral curves.

3) Time Comparison: From Tables II and III, compared with the data-driven methods, our algorithm cannot outperform the supervised algorithms HSID-CNN and TS3C, and the plug-and-play method FastHyMix because they do not need

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

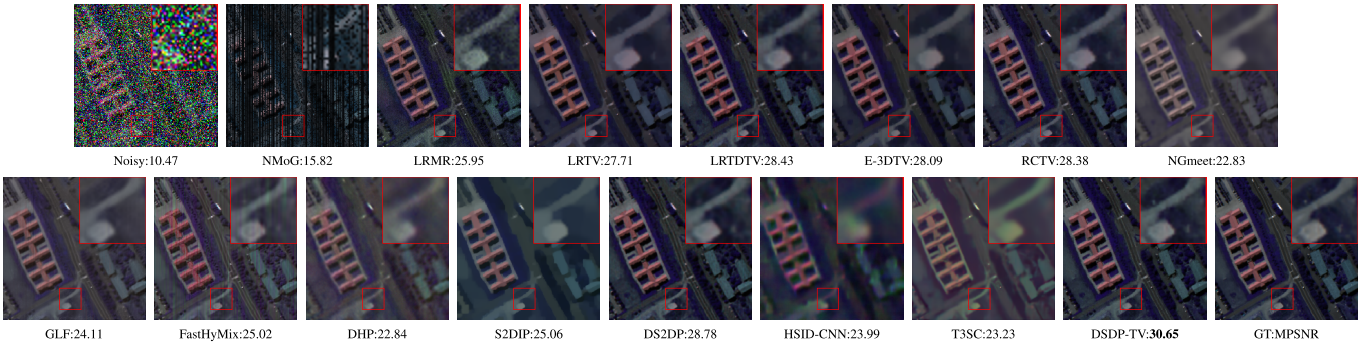


Fig. 4. Recovered images under case (f) of all the competing methods with bands 27-43-62 (R-G-B) of the *PaviaU* dataset. The best MPSNR is highlighted in **bold**.

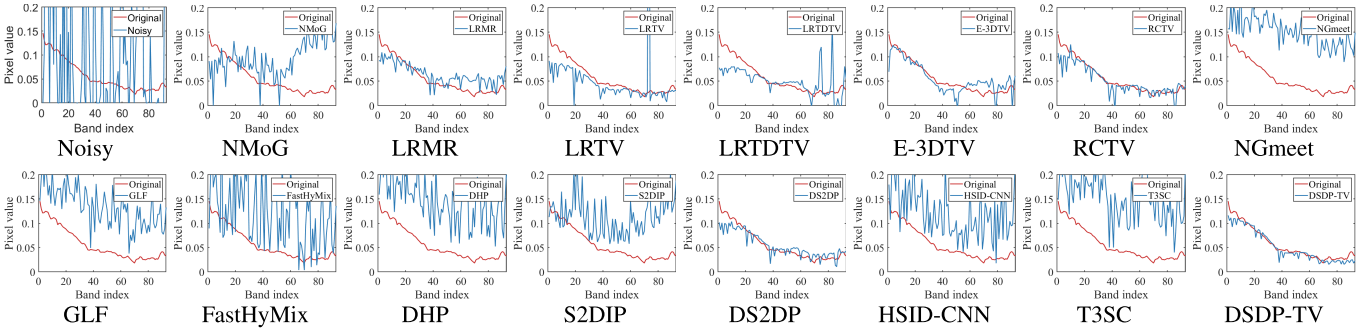


Fig. 5. Recovered spectral curves under case (f) of all the competing methods at point (110, 60) of the *PaviaU* dataset.

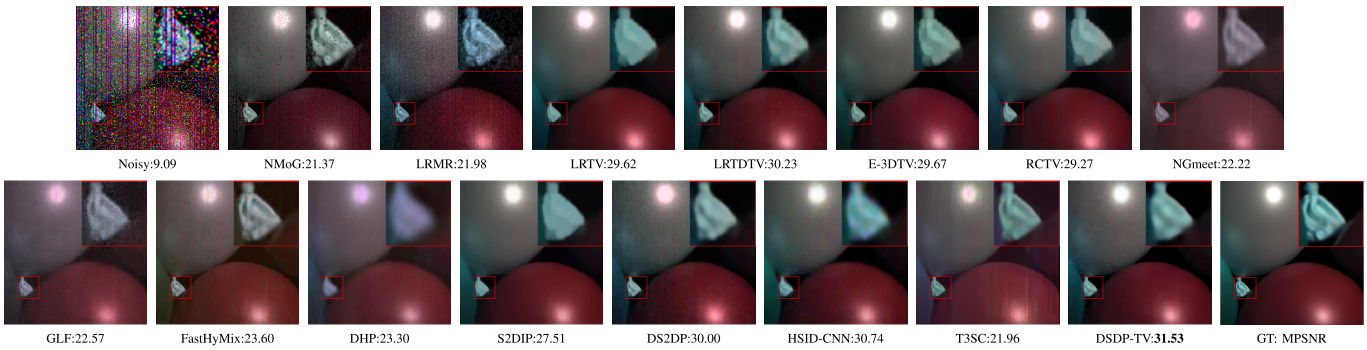


Fig. 6. Recovered images under case (f) of all the competing methods with bands 12-23-29 (R-G-B) of the *balloons* dataset. The best MPSNR is highlighted in **bold**.

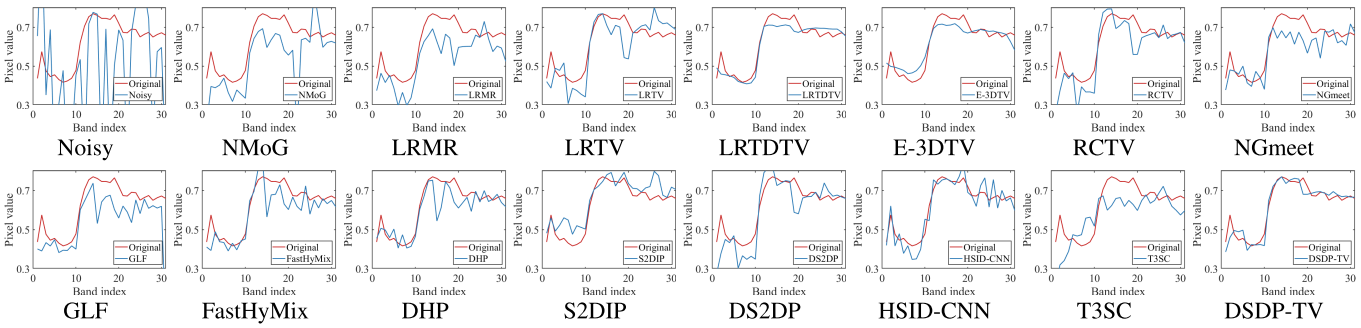


Fig. 7. Recovered spectral curves under case (f) of all the competing methods at point (30, 80) of the *balloons* dataset.

to train the network during inference. However, our algorithm is still relatively fast among DIP-based unsupervised methods. In contrast to DHP and S2DIP, our method represents superior results since the size of the factors to be solved is greatly reduced by subspace-based modeling. Furthermore, our method is also competitive against the similar subspace-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

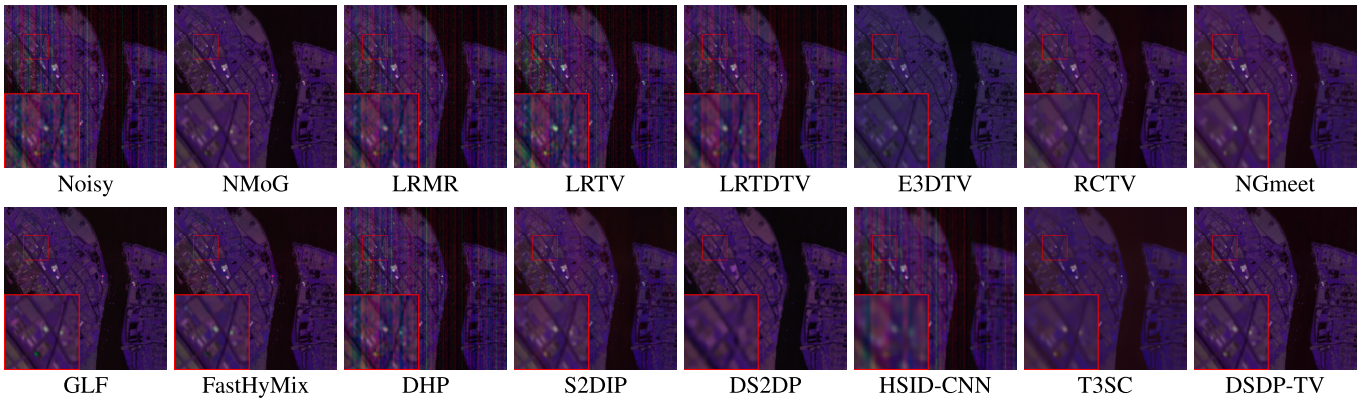


Fig. 8. Recovered images of all the competing methods with bands 155-154-21 (R-G-B) of the *GF-Shanghai* dataset.

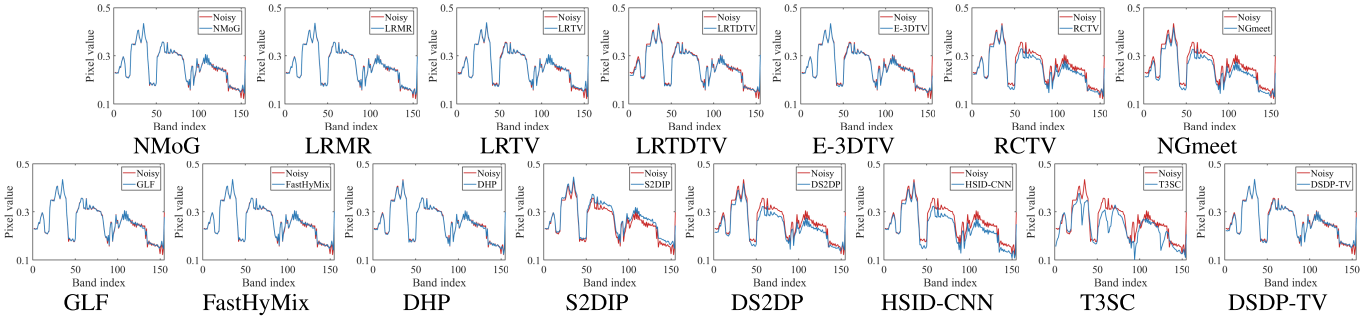


Fig. 9. Recovered spectral curves of all the competing methods at point (200, 250) of the *GF-Shanghai* dataset.

method DS2DP, because the network parameters of DS2DP linearly increase with rank, while the network parameters of our methods remain stable.

The above simulation experiments on HSI and MSI fully exhibit the comparability of our proposed model. Our method exceeds a group of comparative methods in most cases in terms of performance metrics, visual perception, and time costs. The elegant double subspace factor neural networks automatically learn deep priors from single HSI data in an unsupervised learning mode under the LRDF framework. The powerful TV regularity increases the robustness of the model to cope with complex noise. Thus DSDP-TV model demonstrates impressive achievement.

B. Real Data Experiments

To further verify the adaptability of our proposed method on real noise scenes, we executed denoising experiments on two real-world HSI datasets *GF-Shanghai* and *Indian pines*. Parts of the spectra of the two HSIs are heavily polluted by atmosphere, water, and some other physical factors, bringing in severe complex noise, including impulse noise, stripe noise, and deadlines. Given the lack of ground truth, it is impractical to quantitatively evaluate all competing methods. Instead, we qualitatively assess the denoising results by showcasing the reconstruction images as pseudo-RGB images of severely affected bands and displaying the spectral curve recovery. To understand the impact of denoising on downstream tasks, we also conducted HSI classification experiments using the reconstructed *Indian pines* dataset.

1) Visual Comparison: The recovery of the heavily damaged bands is shown in Figs. 8 and 10. From these recovered images, it can be seen that all the competing methods could deal with real noise to some extent, but our method recovers the sharpest and clearest edge profiles. In addition, as is seen from the local zoomed-in region, our proposed method can effectively cope with impulse noise and stripe noise meanwhile keeping spatial structure, which is hard to accomplish for some conventional methods, especially those that do not take full advantage of local smoothness of HSI.

Besides the image spatial structure, an even more important perspective for evaluating denoising algorithms is their ability to preserve the underlying trend of the original spectral curve. Generally, a good denoising algorithm can stably protect certain prior properties of the original spectral curve, such as local continuity, even when the spectral curve exhibits outliers due to complex noise. As can be seen from Figs. 9 and 11, our reconstructed spectral curves show great local continuity in the seriously affected bands, while it keeps a similar trend with the reference spectral curves in the rarely affected bands. This demonstrates the efficiency of our algorithm.

2) Influence on Downstream Applications: The performance of the denoising algorithm greatly affects the quality of the HSI, which further affects the performance on downstream tasks. We compare competing methods by their performance on the HSI classification task. A basic classifier support vector machine (SVM) is applied to execute classification experiments on *Indian pines* dataset after denoising. The classification ground-truth of *Indian pines* dataset is labeled

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

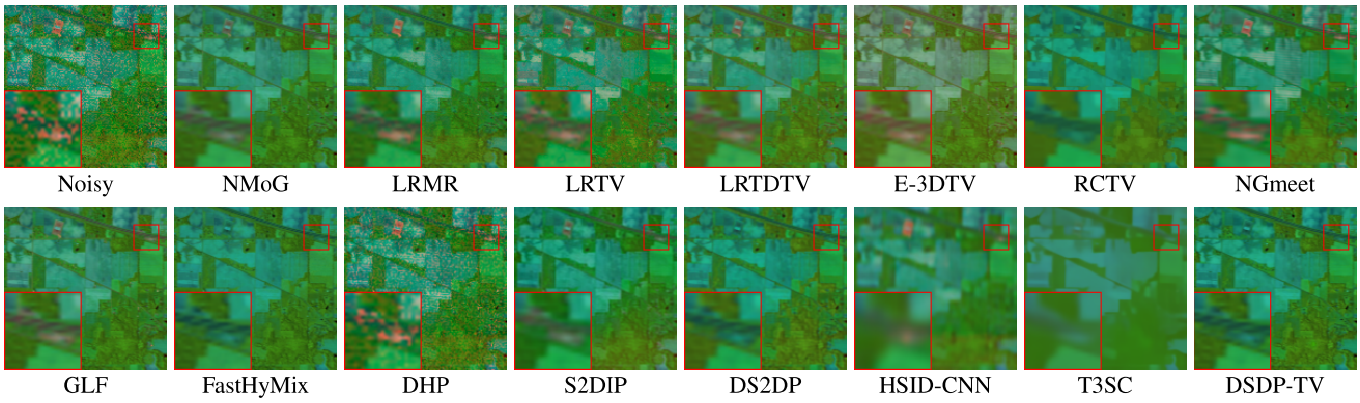


Fig. 10. Recovered images of all the competing methods with bands 1-100-200 (R-G-B) of the *Indian pines* dataset.

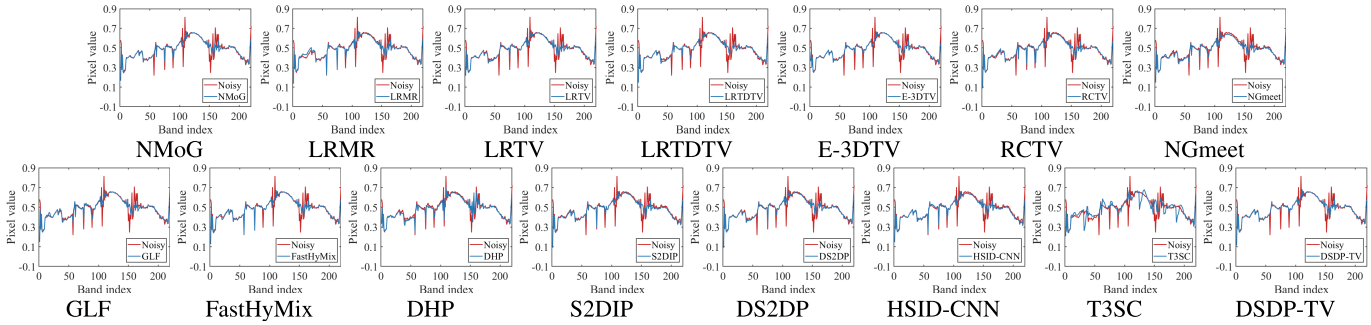


Fig. 11. Recovered spectral curves of all the competing methods at point (90, 60) of the *Indian pines* dataset.

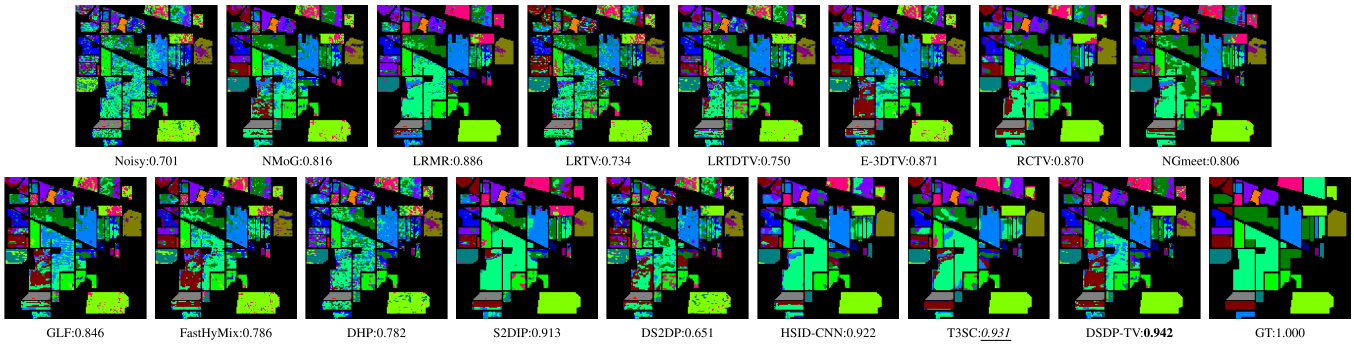


Fig. 12. Classification maps via SVM of all the competing methods of *Indian pines*. Different colors refer to different classification regions. The best and the second best results are highlighted in **bold** and *italic underline*, respectively.

with 16 categories, and the sample size of categories ranges from 20 to 2455. 10% of the samples in each category are randomly selected as the training set, while remaining is the testing set. The quantitative indicator average accuracy (AA) [62] is used to evaluate the classification results. The specific classification maps and indicators are shown in Fig. 12. As seen in Fig. 12, the best accuracy of our method between all the comparison methods proves the effectiveness of our proposed preprocessing denoising method on downstream applications.

V. Discussion

A. Ablation Study

As for our proposed DSDP-TV model, two core modules remarkably matter, that is, the non-local self-similarity attention block in $\mathcal{U}$-branch network to extract long-range spatial self-similarity for the abundance map, and the TV regularizer restriction to enhance network robustness. We adopt an ablation study on the proposed DSDP-TV model with and without these two associated modules to verify their influence. We conduct experiments on the following four models: 1) DSDP-TV without NSAB and TV; 2) DSDP-TV without NSAB; 3) DSDP-TV without TV; and 4) DSDP-TV. The experiment is conducted on *balloons* dataset in the mixed noise case (c) with the same experimental settings and training strategies as Section IV-A. The experimental results are displayed in Table IV.

According to the results in Table IV, we find that the "DSDP-TV wo. NSAB&TV" model has achieved MPSNR = 34.27 db and MSSIM = 0.93, which has already outperformed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY ON BALLOONS DATASET UNDER CASE (c) FOR NSAB MODULE AND TV MODULE

Models	Parameters	MPSNR	MSSIM
DSDP-TV wo. NSAB&TV	2511332	34.2707	0.9289
DSDP-TV wo. NSAB	2511332	36.1150	0.9504
DSDP-TV wo. TV	2765752	35.8317	0.9406
DSDP-TV	2765752	37.2631	0.9632

TABLE V
ABLATION STUDY ON BALLOONS DATASET UNDER CASE (e) BETWEEN DSDP-TV AND S2DIP

Models	DSDP-TV	S2DIP
Original model	35.8646	31.3209
Original model wo. TV	34.3343	31.1400
Original model wo. SSTV	-	30.7726
Original model wo. TV&SSTV	-	30.1984

most of the comparative methods according to Table II. Because the underlying convolutional network elaborately models the spatial–spectral deep prior by our well-designed network structure, and under the framework of the LRDF, the two sub-networks effectively utilize the prior information from two subspace factors. If we add the TV term to "DSDP-TV wo. NSAB&TV," the MPSNR significantly has a gain of 1.84 db, which makes it achieve state-of-the-art already in this experimental case. Because the TV regularization reasonably improves the robustness of the network, especially in mixed noise cases. If we add NSAB to "DSDP-TV wo. NSAB&TV," we find that there is also a gain of 1.56 db for MPSNR. It indicates that the spatial attention block indeed captures rewarding deep features between different regions, which is an essential part of the subspace factor neural network. Finally, if we use the proposed DSDP-TV model, MPSNR can further enhance, reaching 37.26 db, which reveals the joint effectiveness of the NSAB block and TV regularizer to handle complex noise removal in our proposed DSDP-TV model.

To support deep prior for image reconstruction, S2DIP also utilizes TV regularization to encode spatial local smoothness of raw HSI, but it still relies on spatial–spectral constraints delivered by SSTV. Here, we simply compare the effectiveness of the TV regularization of the two methods through an ablation study on *balloons* dataset under case (e) as Table V. From Table V, we can see that DSDP achieves better performance even without TV regularization. In contrast, the performance of S2DIP is significantly constrained even with TV regularization. This suggests that using a dual-branch DIP network under LRDF is superior to using a single DIP network.

B. Parameter Analysis

In the proposed model DSDP-TV, four parameters need to be pre-specified, that is, r, w, α, and β. We then analyze the effects of these four parameters.

The rank r mainly characterizes the spectral correlation of HSI, and the more accurate rank can be estimated by the HySime algorithm [63]. We know that the number of parameters of the network is related to the rank; therefore,

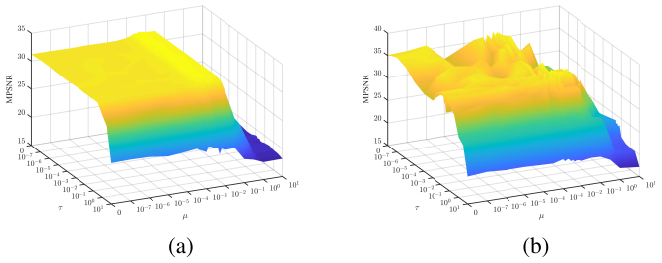


Fig. 13. Sensitivity analysis of the parameters τ, μ under case (c). (a) *PaviaU* dataset under case (c). (b) *Balloons* dataset under case (c).

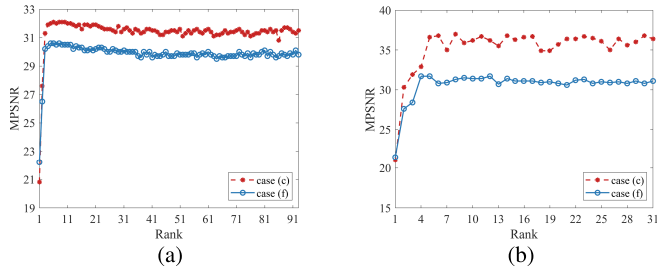


Fig. 14. Sensitivity analysis of the rank r under case (c) and case (f). (a) *PaviaU* dataset. (b) *Balloons* dataset.

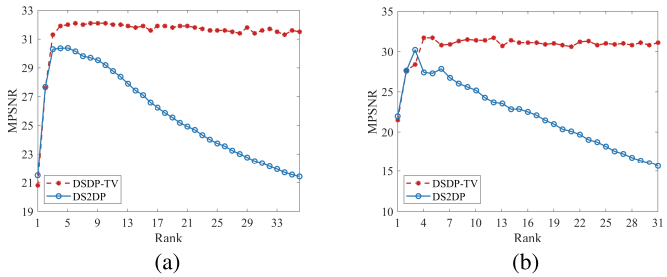


Fig. 15. Sensitivity analysis of the rank r for DS2DP and DSDP-TV. (a) *PaviaU* dataset under case (c). (b) *Balloons* dataset under case (f).

we provide a sensitivity analysis of the denoising performance with different rank r in Fig. 14. From Fig. 14, we can find that the DSDP-TV model exhibits stability and great performance within a certain range rank under case (c) and case (f) on *PaviaU* dataset and *balloons* dataset. The denoising performance can achieve its peak at a small rank setting and remains stable or slightly decreases with the increasing rank. This reveals the spectral low-rank properties for hyperspectral data, and that the over-parameterized network remains robust to unknown rank and still can reach the global optimum point on the optimization process.

In addition compared to the similar subspace-based self-supervised denoising method DS2DP, DSDP-TV exhibits greater robustness to the changing rank. Fig. 15 exhibits the comparison analysis of the two methods under different rank r. From Fig. 15, we can find that, with uncertain rank estimation, the over-parameterized network structure of DSDP-TV still maintains a superior stable denoising effect, while DS2DP represents a huge shake.

w indicates the sliding window size of the window-based spatial attention module, which balances the computational complexity and feature extraction capabilities of self-attention operation. Generally, the window size w should be kept

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
SENSITIVITY ANALYSIS OF w. THE BEST IS HIGHLIGHTED IN **BOLD**

Noise cases\w	**4**	**5**	**6**	**8**	**10**	**12**
PaviaU (case (e))	34.01	34.08	**34.13**	33.87	33.82	34.01
PaviaU (case (f))	30.66	30.58	30.59	**30.77**	30.42	30.28
balloons (case (e))	35.88	35.88	35.83	35.79	**36.36**	35.97
balloons (case (f))	31.61	31.58	31.43	**31.48**	31.20	31.38

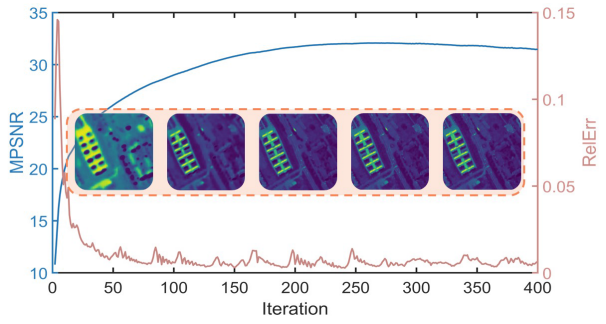


Fig. 16. History of MPSNR and RelError of denoising process.

at a minimum under acceptable denoising performance. We analyze the denoising performance with different w under case (e) and case (f) on *PaviaU* dataset and *balloons* dataset in Table VI. From Table VI, we find that DSDP-TV shows a certain robustness to the value of w; thus we highly recommend selecting w between 6 and 10.

The parameters α and β balance the importance of $\|\mathcal{U}\|_{\mathrm{TV}}$ and $\|\mathbf{V}\|_{\mathrm{TV}}$ regularity, while they further control the effect of regularizer term on the parameterized network. The denoising performance of the model is closely related to such two parameters, so we analyze the sensitivity of these two parameters under case (c) on *PaviaU* dataset and *balloons* dataset in Fig. 13. Considering offsetting the effect of the uneven sizes of $\mathcal{U}$ and $\mathbf{V}$, we execute division operators $\alpha := (\tau/MNr) and \beta := (\mu/pr)$, and display the relationship between MPSNR and the re-scaled parameters in the intervals of $\tau \in [0, 10], \mu \in [0, 10]$. From Fig. 13, we find that the algorithm exhibits different performances at different parameter settings and that the denoising effect significantly improves at a certain range of parameter settings, which guides us to fine-tune the parameters in practice. In fact, we only simply fine-tune τ at $[10^{-2}, 10^{-1}]$, and μ at $[10^{-3}, 10^{-2}]$.

C. Convergence Analysis

To verify the convergence of the proposed DSDP-TV model and the rationality of the convergence conditions RelError, we visualize the changes in MPSNR and RelError of the recovered HSI of *PaviaU* dataset under case (c) in 400 iteration steps as Fig. 16. From Fig. 16, it shows a steady increase in MPSNR at the first period of the iterations, followed by a slight decrease, which is similar to the convergence process of the comparable unsupervised methods S2DIP [41] and DS2DP [38]. Meanwhile, the value of RelError decreases dynamically over the iterations until MPSNR reaches its highest point, which is consistent with what is described in S2DIP [41]. Therefore, the proposed method can accurately divide the noise with the clean HSI on the denoising process and achieve convergence under gentle conditions.

VI. CONCLUSION AND FUTURE WORK

This article introduces a generic DSDP-TV model for removing mixed noise from hyperspectral imaging (HSI). The method utilizes prior knowledge from the original HSI and applies it to subspace factors within the LRDF. Since subspace methods already encode the spectral low-rankness of HSI data, fully prior encoding can be achieved by only characterizing spatial priors on subspace factors. To accomplish this, we design two sub-network structures based on convolutional neural networks and attention mechanisms to parameterize the entire subspace factors, exploring their deep priors. Furthermore, we integrate classical TV regularization into the optimization process to enhance network reconstruction and model robustness. Parameters within the neural network can be learned unsupervised from HSI observations. Extensive experimental results validate the effectiveness of this method.

In the future, we will further explore prior information on subspace factors by integrating more advanced regularization techniques and network modules. Additionally, we aim to extend the DSDP-TV model to other applications such as HSI inpainting and super-resolution.

REFERENCES

[1] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.
[2] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2022.
[3] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.
[4] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.
[5] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.
[6] W. He, M. Wang, Y. Chen, and H. Zhang, "An unsupervised dehazing network with hybrid prior constraints for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514715.
[7] L. Zhang, W. Wei, Y. Zhang, F. Li, C. Shen, and Q. Shi, "Hyperspectral compressive sensing using manifold-structured sparsity prior," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3550–3558.
[8] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.
[9] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009.
[10] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.
[11] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1–37, 2011.
[12] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2013.
[13] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis with a new tensor nuclear norm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 925–938, Apr. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

[14] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2949–2956.

[15] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[16] H. Zhang, L. Liu, W. He, and L. Zhang, "Hyperspectral image denoising with total variation regularization and nonlocal low-rank tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3071–3084, May 2019.

[17] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev. Soc. Ind. Appl. Math.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.

[18] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2015.

[19] Y. Wang, J. Peng, Q. Zhao, Y. Leung, X.-L. Zhao, and D. Meng, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.

[20] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C.-W. Chan, and W. Philips, "Spatial-spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[21] Q. Xie, Q. Zhao, D. Meng, and Z. Xu, "Kronecker-basis-representation based tensor sparsity and its applications to tensor recovery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 8, pp. 1888–1902, Aug. 2018.

[22] J. Xue, Y. Zhao, W. Liao, and J. C. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[23] L. Zhuang, X. Fu, M. K. Ng, and J. M. Bioucas-Dias, "Hyperspectral image denoising based on global and nonlocal low-rank factorizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10438–10454, Dec. 2021.

[24] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 6868–6877.

[25] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.

[26] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[27] J. Peng et al., "Fast noise removal in hyperspectral images via representative coefficient total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546017.

[28] J. Peng et al., "Learnable representative coefficient image denoiser for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506516.

[29] X. Cao, X. Fu, C. Xu, and D. Meng, "Deep spatial–spectral global reasoning network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504714.

[30] K. Wei, Y. Fu, and H. Huang, "3-D quasi-recurrent neural network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 363–375, Jan. 2021.

[31] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Sep. 2018.

[32] Y. Chang, L. Yan, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2018.

[33] F. Xiong, J. Zhou, S. Tao, J. Lu, J. Zhou, and Y. Qian, "SMDS-net: Model guided spectral–spatial network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 5469–5483, 2022.

[34] T. Bodrito, A. Zouaoui, J. Chanussot, and J. Mairal, "A trainable spectral–spatial sparse coding model for hyperspectral image restoration," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 5430–5442.

[35] L. Zhuang and M. K. Ng, "FastHyMix: Fast and parameter-free hyperspectral image mixed noise removal," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4702–4716, Aug. 2021.

[36] T. X. Jiang et al., "Adaptive hyperspectral mixed noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5511413.

[37] B. Rasti, B. Koirala, P. Scheunders, and P. Ghamisi, "UnDIP: Hyperspectral unmixing using deep image prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504615.

[38] Y.-C. Miao, X.-L. Zhao, X. Fu, J.-L. Wang, and Y.-B. Zheng, "Hyperspectral denoising using unsupervised disentangled spatiospectral deep priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513916.

[39] Q. Zhang, Q. Yuan, M. Song, H. Yu, and L. Zhang, "Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 6356–6368, 2022.

[40] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[41] Y.-S. Luo, X.-L. Zhao, T.-X. Jiang, Y.-B. Zheng, and Y. Chang, "Hyperspectral mixed noise removal via spatial–spectral constrained unsupervised deep image prior," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9435–9449, 2021.

[42] J. W. P. Sun, H. Li, W. Li, X. Meng, C. Ge, and Q. Du, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Jun. 2021.

[43] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[44] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.

[45] S. Gu, Q. Xie, D. Meng, W. Zuo, X. Feng, and L. Zhang, "Weighted nuclear norm minimization and its applications to low level vision," *Int. J. Comput. Vis.*, vol. 121, no. 2, pp. 183–208, Jan. 2017.

[46] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis: Exact recovery of corrupted low-rank tensors via convex optimization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5249–5257.

[47] B. Rasti, M. O. Ulfarsson, and P. Ghamisi, "Automatic hyperspectral image restoration using sparse and low-rank modeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2335–2339, Dec. 2017.

[48] Y. Chen, X. Cao, Q. Zhao, D. Meng, and Z. Xu, "Denoising hyperspectral image with non-i.i.d. noise structure," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 1054–1066, Mar. 2018.

[49] Y. Chen, Y. Guo, Y. Wang, D. Wang, C. Peng, and G. He, "Denoising of hyperspectral images using nonconvex low rank matrix approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5366–5380, Sep. 2017.

[50] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D, Nonlinear Phenomena*, vol. 60, nos. 1–4, pp. 259–268, 1992.

[51] J. Peng, Y. Wang, H. Zhang, J. Wang, and D. Meng, "Exact decomposition of joint low rankness and local smoothness plus sparse matrices," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5766–5781, May 2023.

[52] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.

[53] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4260–4268.

[54] Y. Zheng, T. Huang, X. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020.

[55] L. Pang, W. Gu, and X. Cao, "TRQ3DNet: A 3D quasi-recurrent and transformer based network for hyperspectral image denoising," *Remote Sens.*, vol. 14, no. 18, p. 4598, Sep. 2022.

[56] Y. Zhao, D. Zhai, J. Jiang, and X. Liu, "ADRN: Attention-based deep residual network for hyperspectral image denoising," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2020, pp. 2668–2672.

[57] Q. Shi, X. Tang, T. Yang, R. Liu, and L. Zhang, "Hyperspectral image denoising using a 3-D attention denoising network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10348–10363, Dec. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.

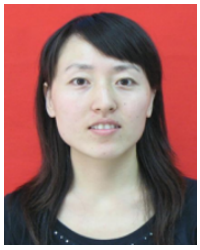
[58] F. Xiong, J. Zhou, Q. Zhao, J. Lu, and Y. Qian, "MAC-net: Model-aided nonlocal neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519414.
[59] M. Li, Y. Fu, and Y. Zhang, "Spatial-spectral transformer for hyperspectral image denoising," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 1, pp. 1368–1376.
[60] H. Wang, J. Peng, X. Cao, J. Wang, Q. Zhao, and D. Meng, "Hyperspectral image denoising via nonlocal spectral sparse subspace representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5189–5203, 2023.
[61] D. Meng and F. De la Torre, "Robust matrix factorization with unknown noise," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 1337–1344.
[62] L. Wang and C. Zhao, *Hyperspectral Image Processing*. Berlin, Germany: Springer, 2016.
[63] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

Kexin Shi received the B.Sc. degree in statistics from Northwestern Polytechnical University, Xi'an, China, in 2022. She is currently pursuing the M.Sc. degree in mathematics with Xi'an Jiaotong University, Xi'an.

Her research interests include image processing and model-driven deep learning.

Jiangjun Peng (Member, IEEE) received the B.Sc. degree from Northwest University, Xi'an, China, in 2015, and the M.Sc. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, in 2018 and 2023, respectively.

In 2017, he participated in an Academic Exchange at Chinese University of Hong Kong, Sha Tin District, he worked as an Assistant Researcher at the Institute of Future Cities, Glasgow, U.K. From 2018 to 2020, he worked as a Researcher at Tencent, Shenzhen, China. Currently, he is an Associate Professor at the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an. His research interests include hyperspectral image processing, low-rank matrix/tensor analysis, and model-driven deep learning.



Jing Gao received the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2007.

She worked as a Post-Doctoral Researcher with the University of Cambridge, Cambridge, U.K., from 2008 to 2009, where she was an Academic Visitor, from 2016 to 2017. She is currently an Associate Professor with the School of Mathematics and Statistics, Xi'an Jiaotong University. Her research interest includes mathematical foundations of artificial intelligence.

Yisi Luo received the B.Sc. degree from the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an, China.

His research interests include image processing, machine learning, and multidimensional data analysis.

He served as a regular reviewer or program committee for CVPR, ICCV, ECCV, AAAI, and ACM MM.

Shuang Xu received the B.Sc. degree from Henan University, Kaifeng, China, in 2016, and the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2021.

Currently, he is an Associate Professor at the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an. His research interests include hyperspectral image processing and multimodal remote sensing image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:19:53 UTC from IEEE Xplore. Restrictions apply.